

Data Science and Machine Learning

DEPLOYMENT ENVIRONMENTS

VIRTUAL ENVIRONMENTS

Virtual Environments

(venv)



Pandas
numpy
Matplotlib
…
Requests
Tensorflow
Mlflow
Sckit-learn

MyComputer

Virtual Environments

(venv)



Pandas=2.22.09
numpy
matplotlib
requests

MyComputer



BeatufiulSoup
Pandas=1.0.19

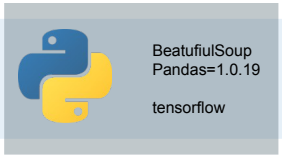
tensorflow

Virtual Environments

(venv)

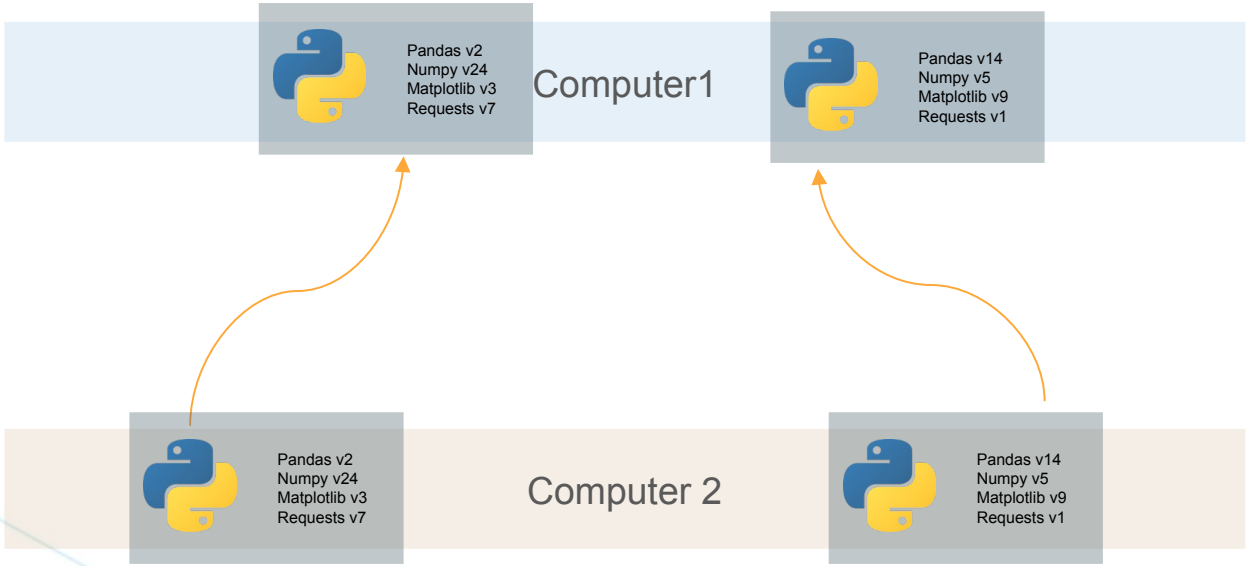


MyComputer



Virtual Environments

(venv)



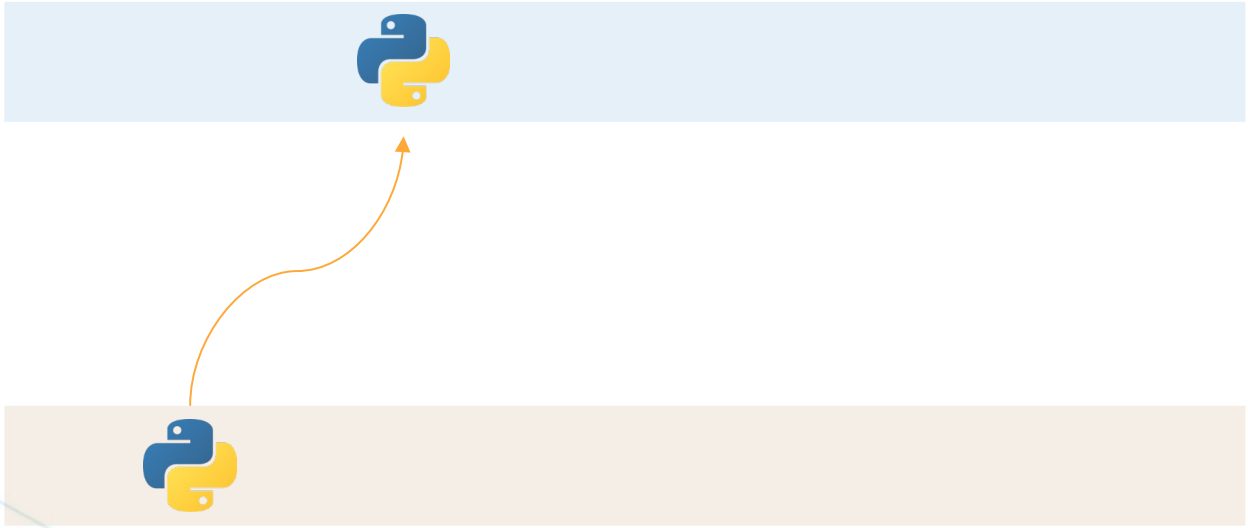
DEMO on VENVS

EXERCISE

02. Virtual Envs.ipynb

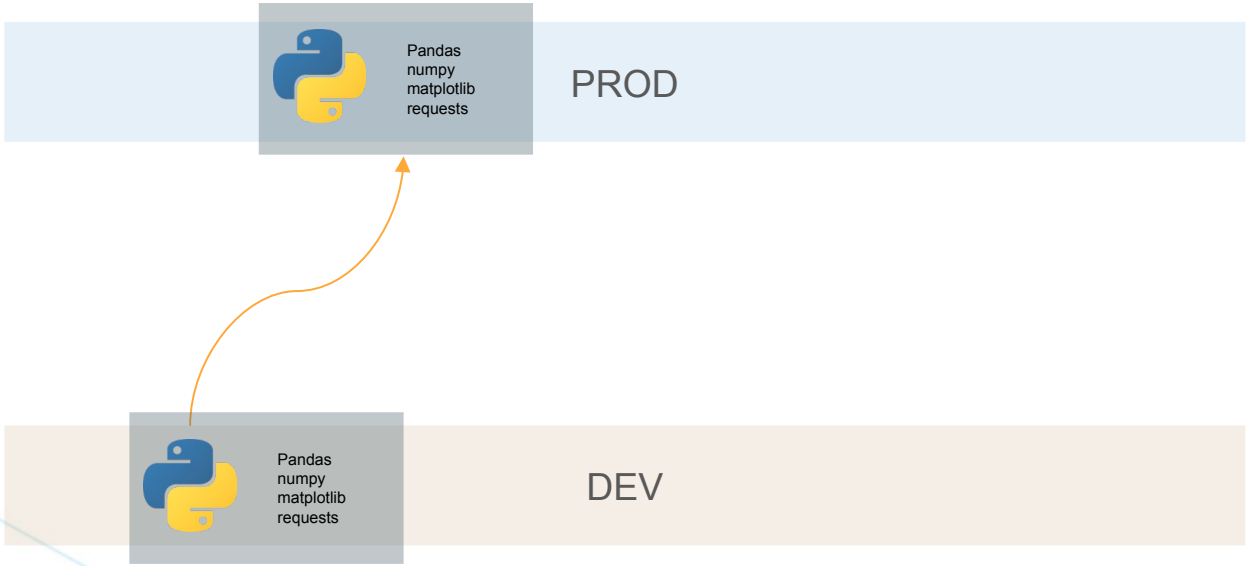
Virtual Environments

(venv)

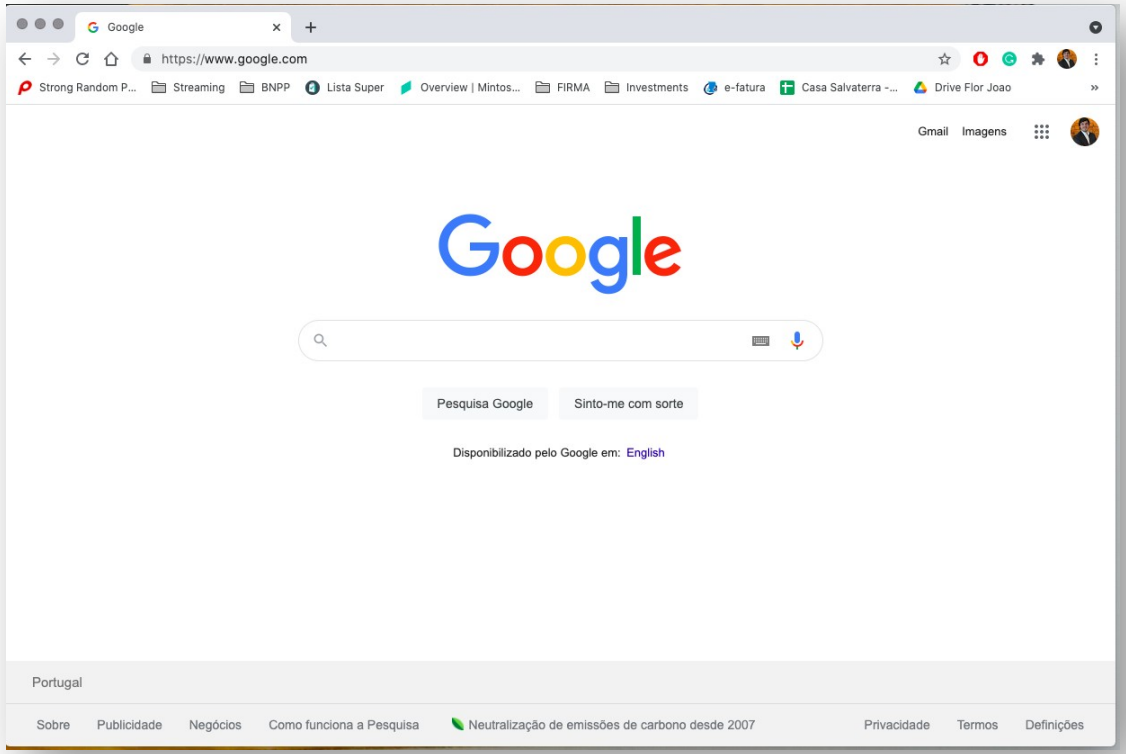


Virtual Environments

(venv)



AM I SEEING DIRECTLY WHAT THE DEVELOPERS ARE



DEV vs PROD

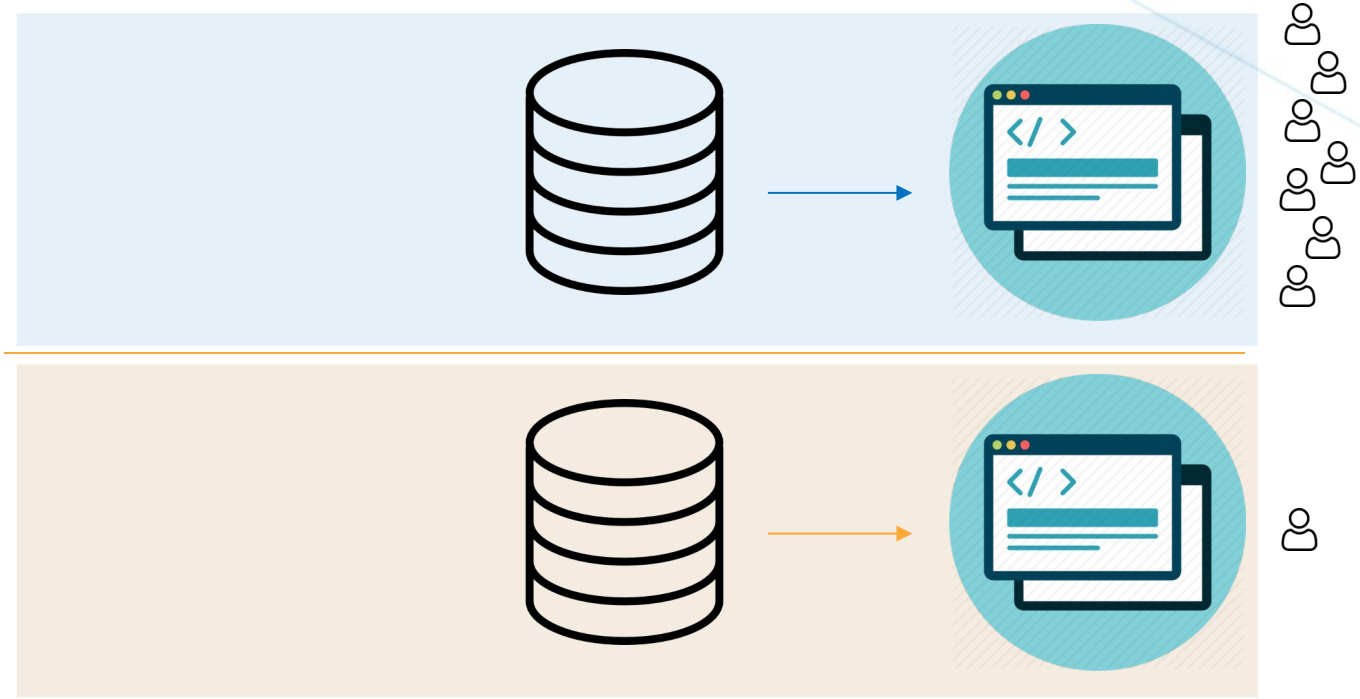


DEV vs PROD

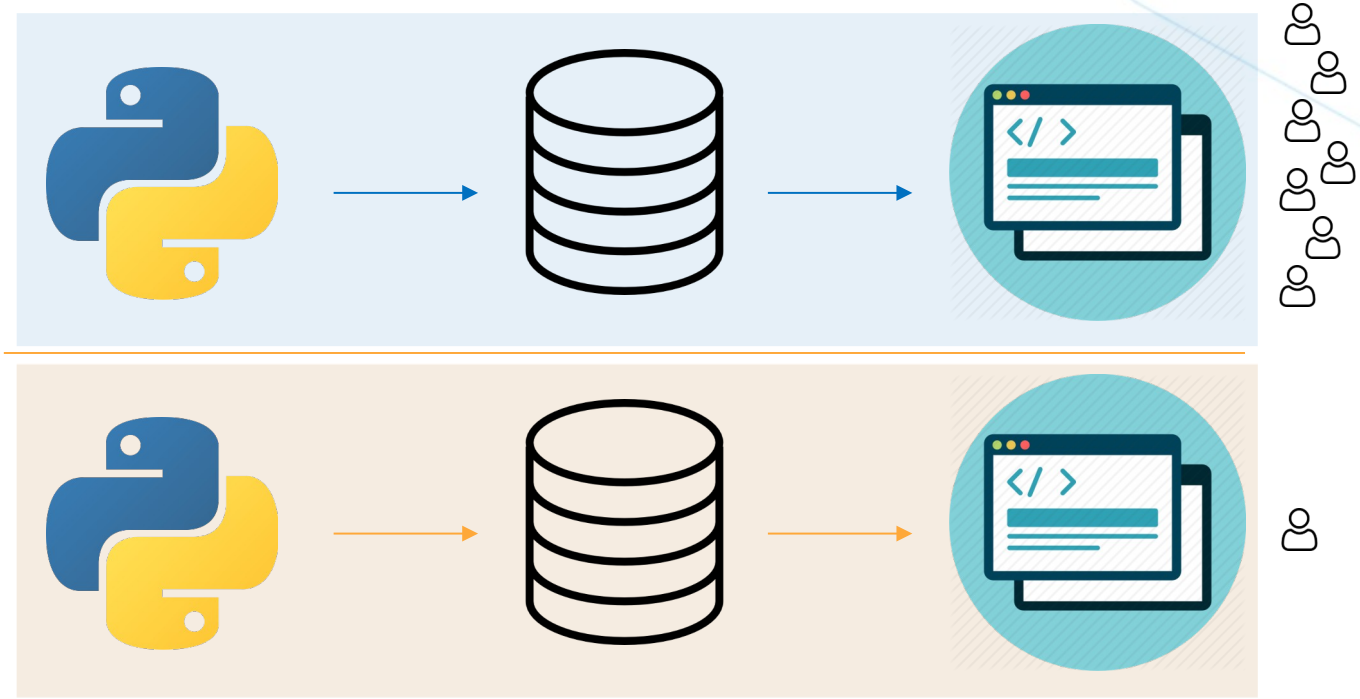




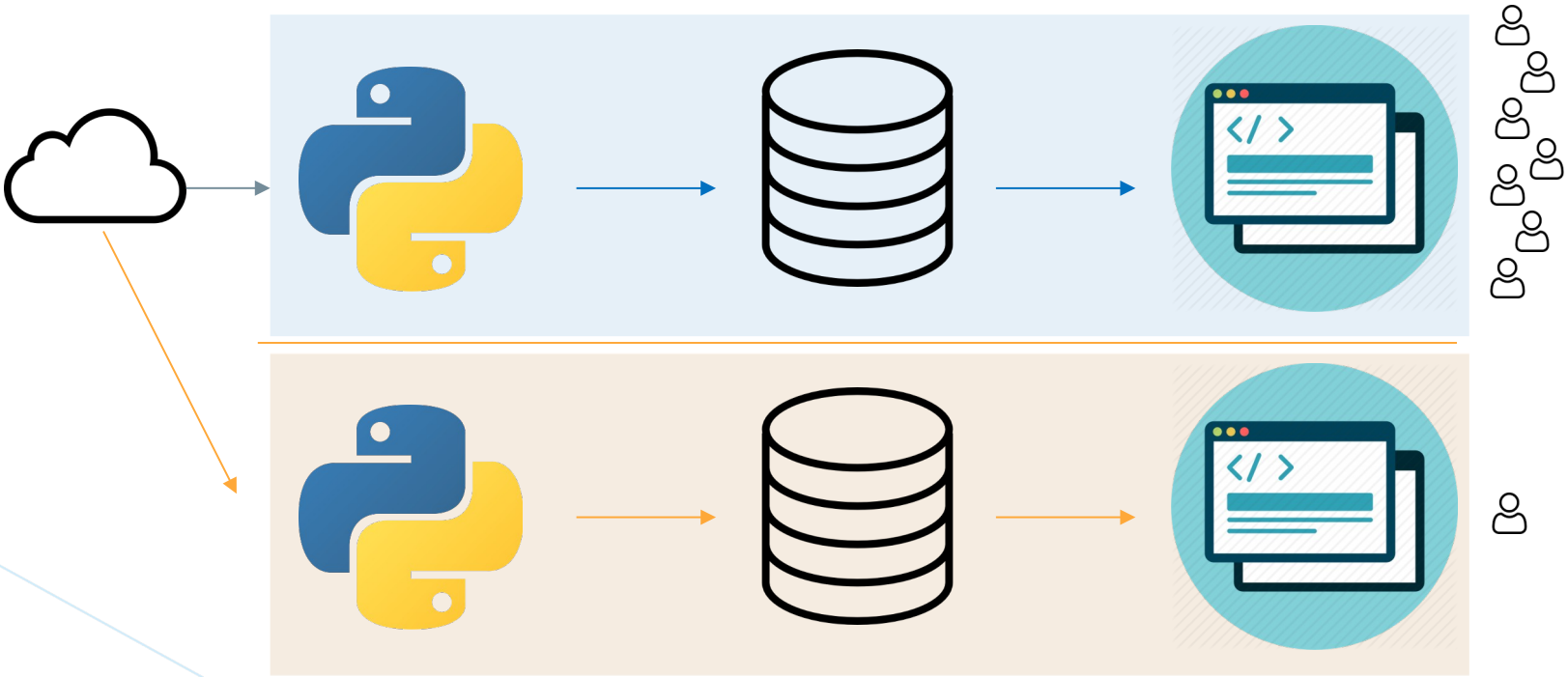
DEV vs PROD



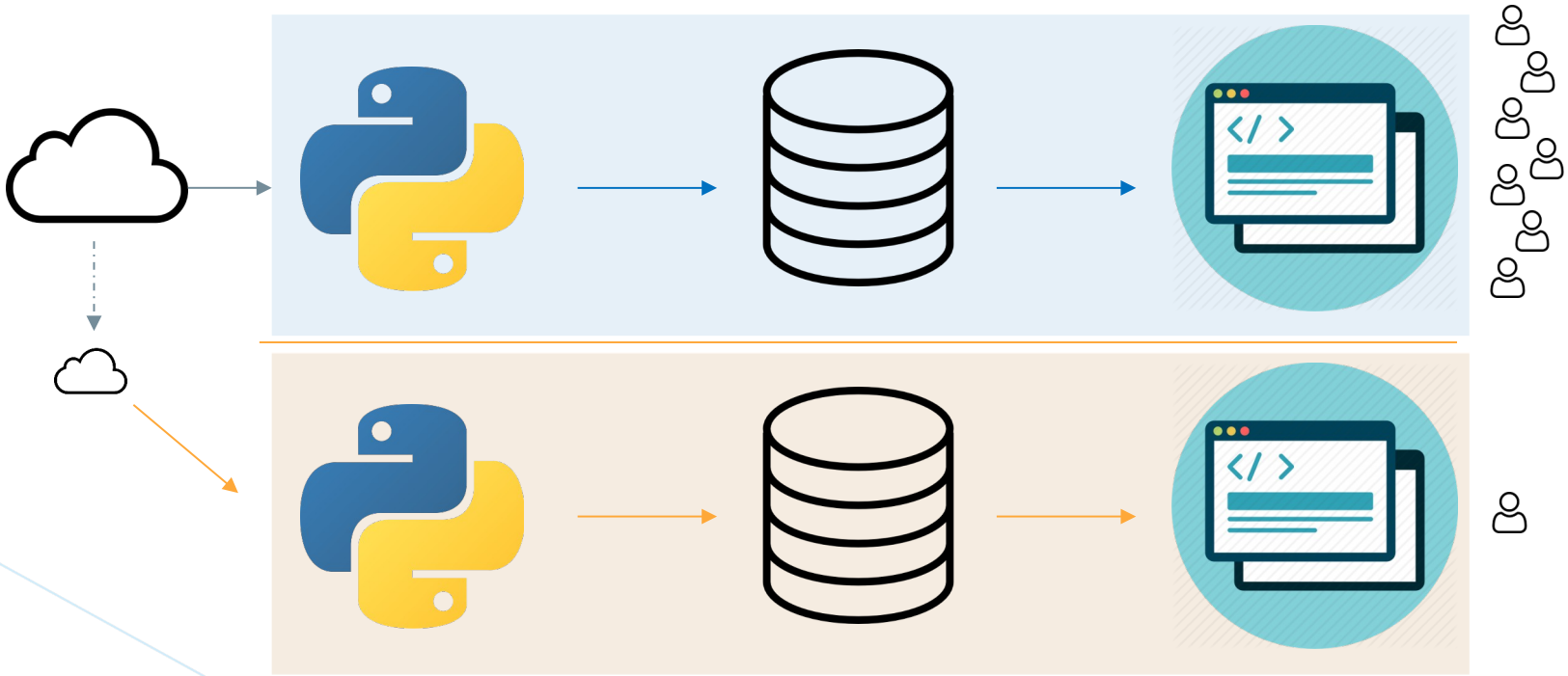
DEV vs PROD



DEV vs PROD

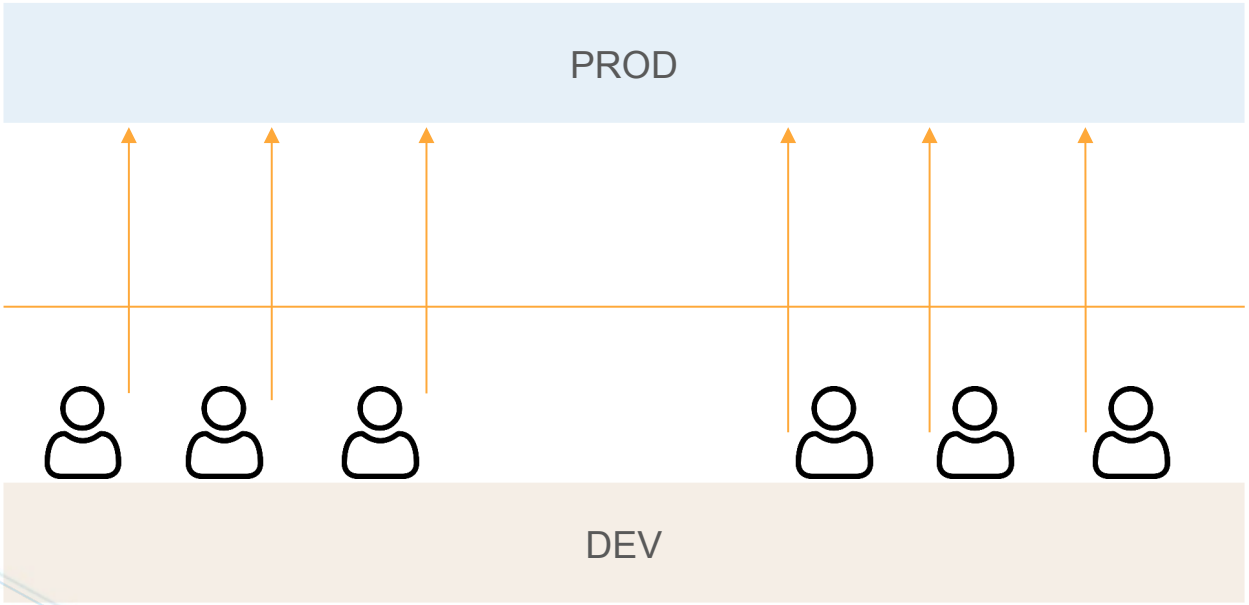


DEV vs PROD

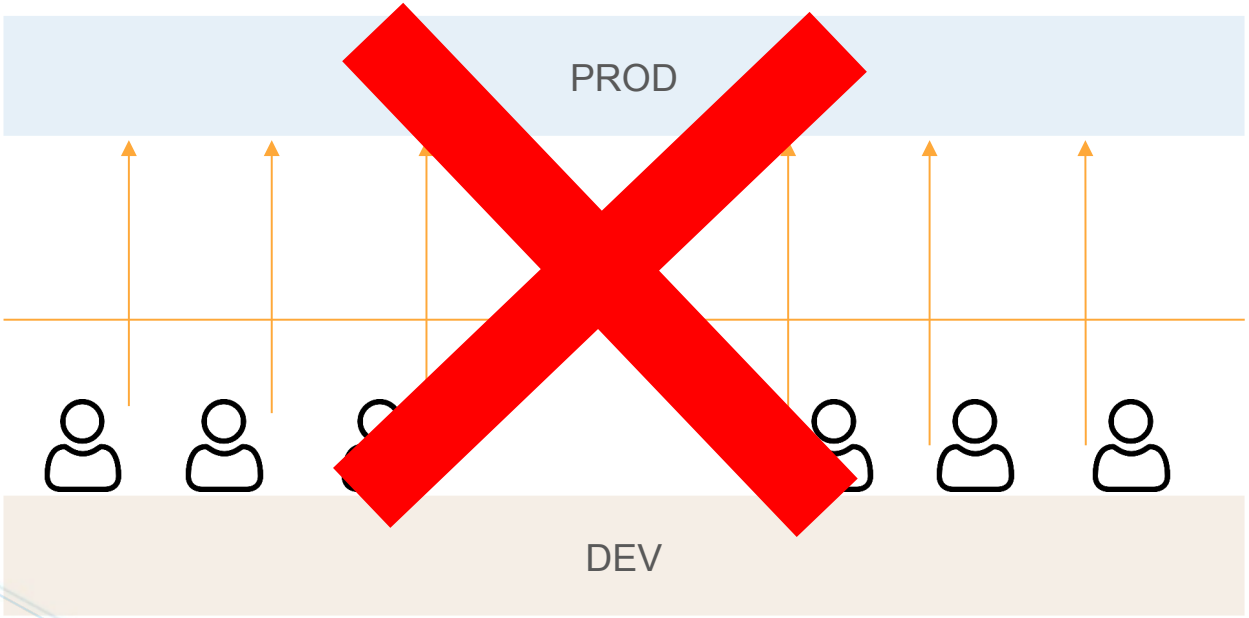


WHAT ABOUT DEPLOYING THINGS FROM ONE ENVIRONMENT TO THE OTHER?

DEV vs PROD

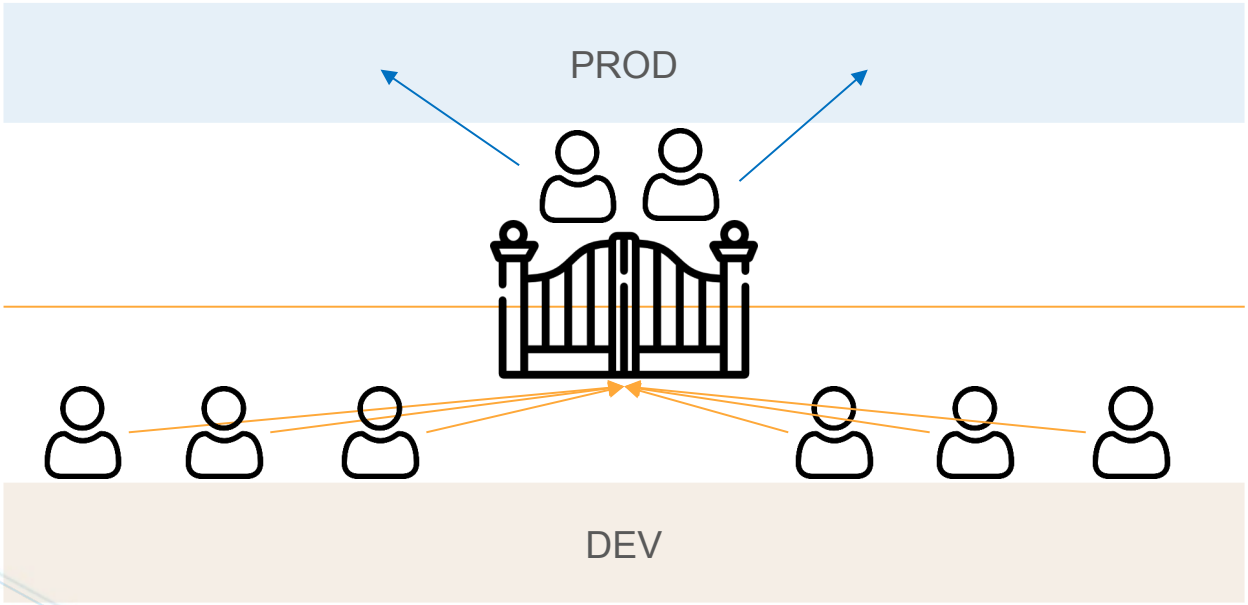


DEV vs PROD



Welcome

DEV vs PROD - Gatekeeping

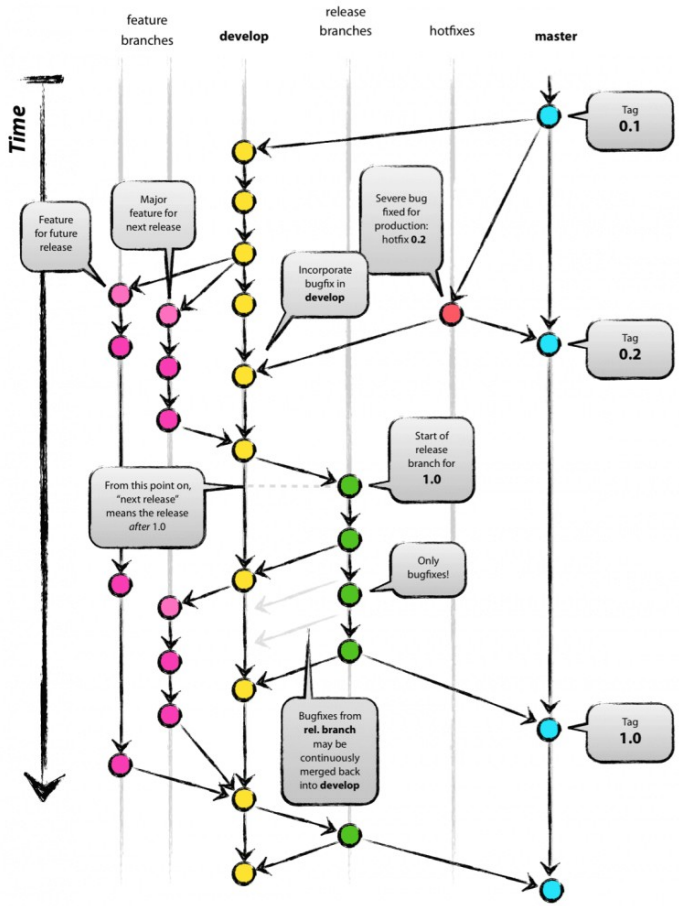


BUT HOW DO WE “GIVE” WHAT WE JUST DID TO THESE PEOPLE?

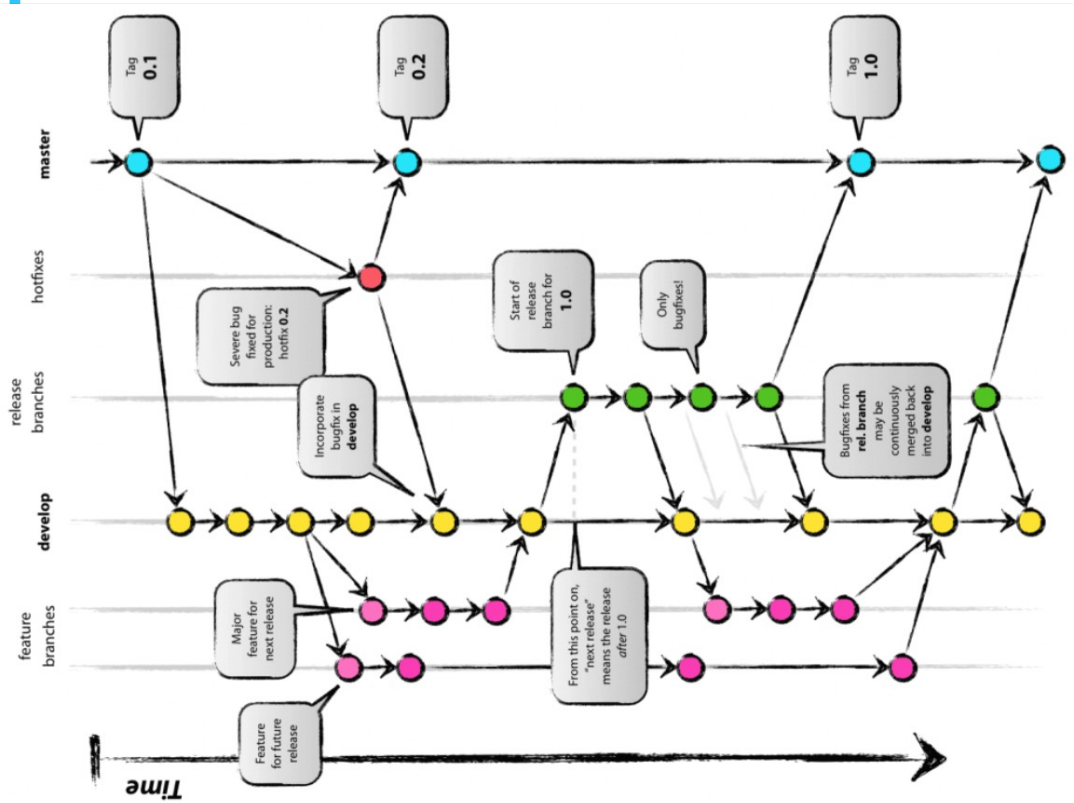
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THERE ARE ACTUALLY A FEW AWAYS

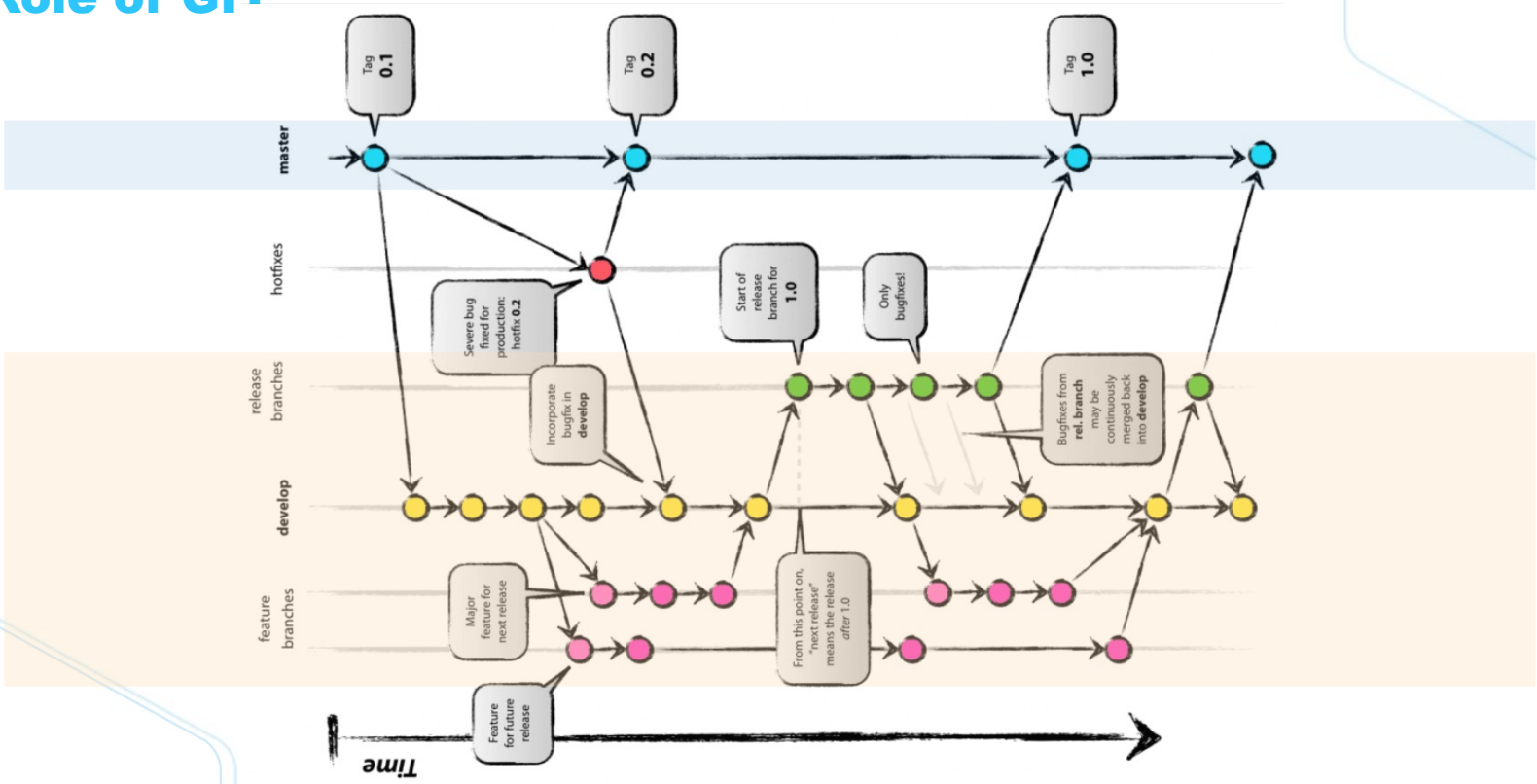
Role of GIT


feature branches
develop
release branches
hotfixes
master
Time
Tag 0.1
Major feature for next release
Feature for future release
Severe bug fixed for production: hotfix 0.2
Incorporate bugfix in develop
Tag 0.2
Start of release branch for 1.0
From this point on, "next release" means the release after 1.0
Only bugfixes!
Bugfixes from rel. branch may be continuously merged back into develop
Tag 1.0

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Role of GIT

PROD







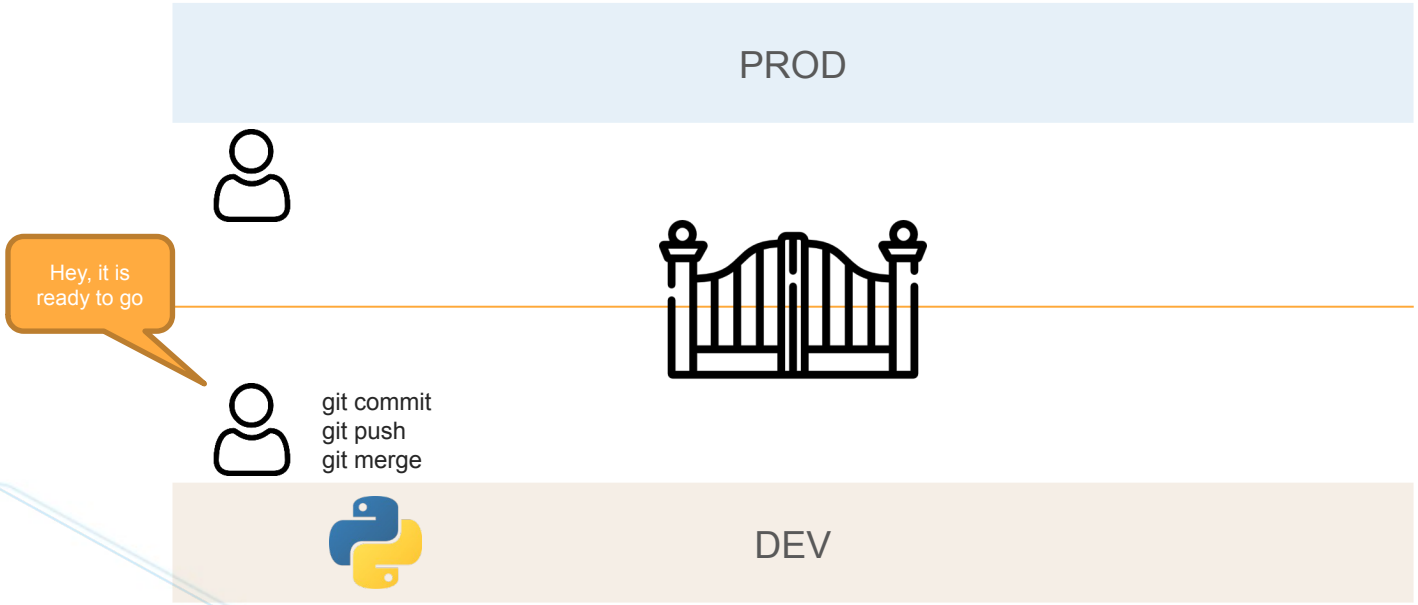
git commit
git push
git merge



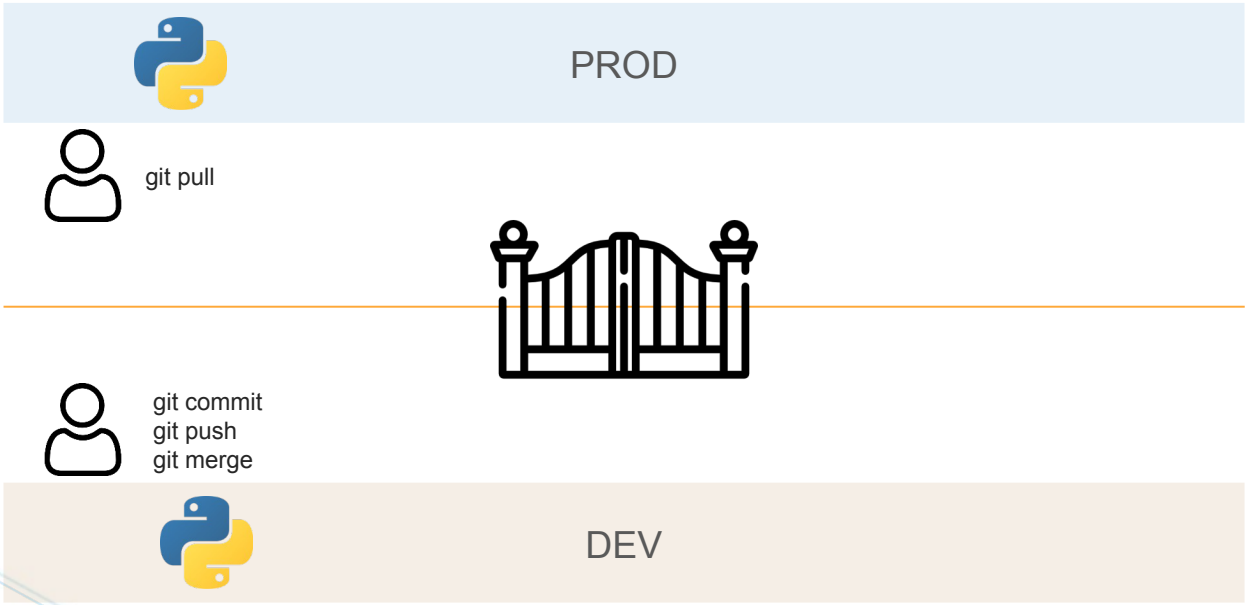
DEV

Welcome

Role of GIT



Role of GIT



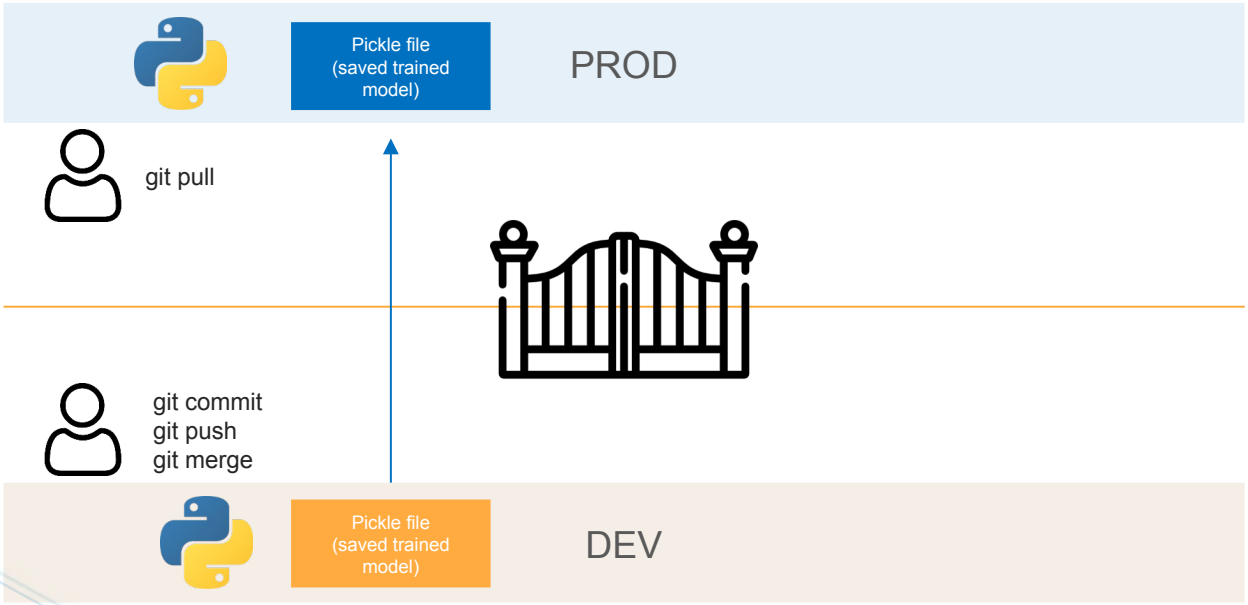
Welcome

HOW ABOUT MACHINE LEARNING MODELS?

HOW ABOUT MACHINE LEARNING MODELS?

We can use pickle files!

Role of GIT


Pickle file
(saved trained
model)
PROD
git pull
git commit
git push
git merge
Pickle file
(saved trained
model)
DEV

BUT... I JUST NEED TO TELL A PERSON:

“PUT THIS IN PROD”... AND HE/SHE DOES IT?

WHAT’S THE POINT THEN?

Role of GIT

PROD



git commit
git push
git merge

DEV

Role of GIT



PROD

CODE REVIEW:

PULL REQUEST

git commit
git push
git merge

DEV

Lets Review

PROD

- More than one environment
- DEVs cant put things in PROD
- Before going to PROD, we need to make a code review

DEV

HAVING THIS IS REALLY GOOD!

HAVING THIS IS REALLY GOOD...

... HOWEVER, QUITE A FEW MANUAL STEPS: CAN WE MAKE IT BETTER?

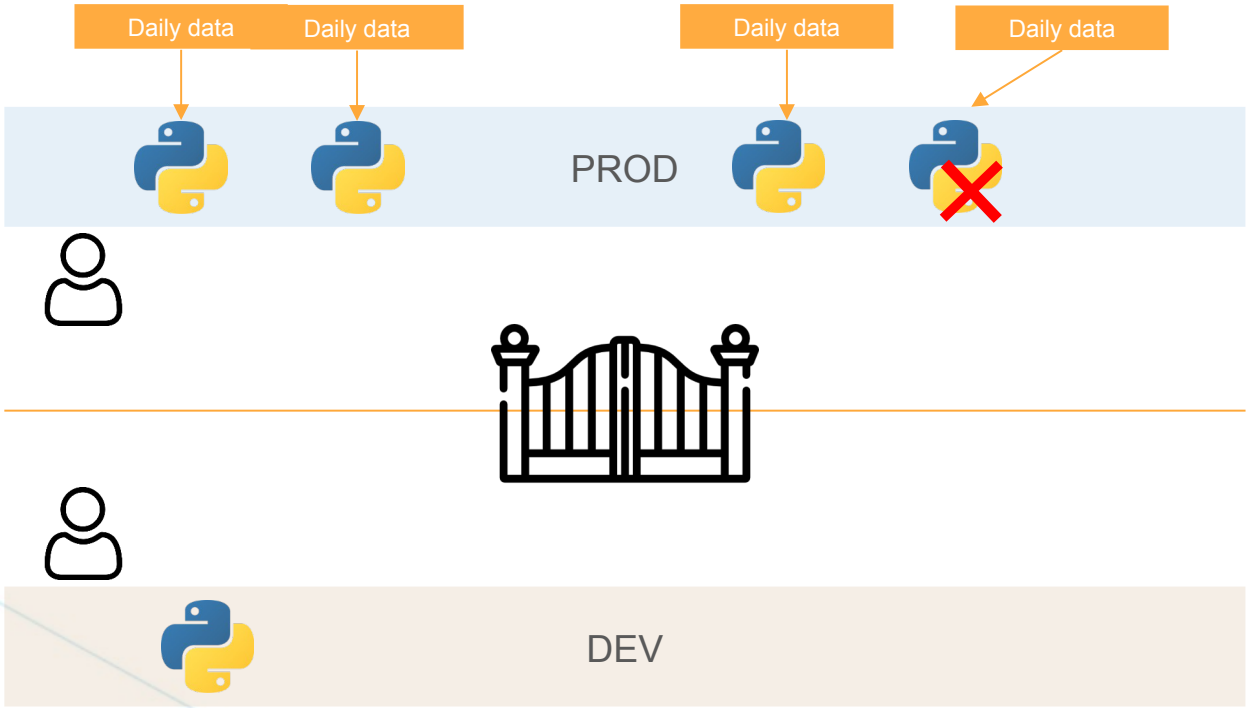
CI CD



CI CD


CI/CD Process
SIGN OFF GIVEN THE RESULTS OF THE TESTS
Commit change
Build
Test
Deliver
Deploy

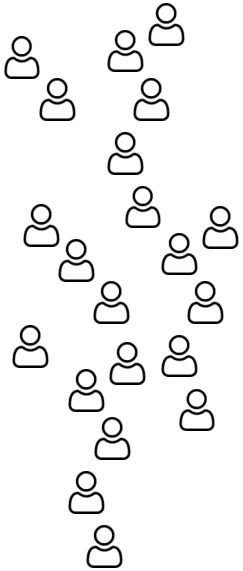
Topics to look out for



- Production Maintenance

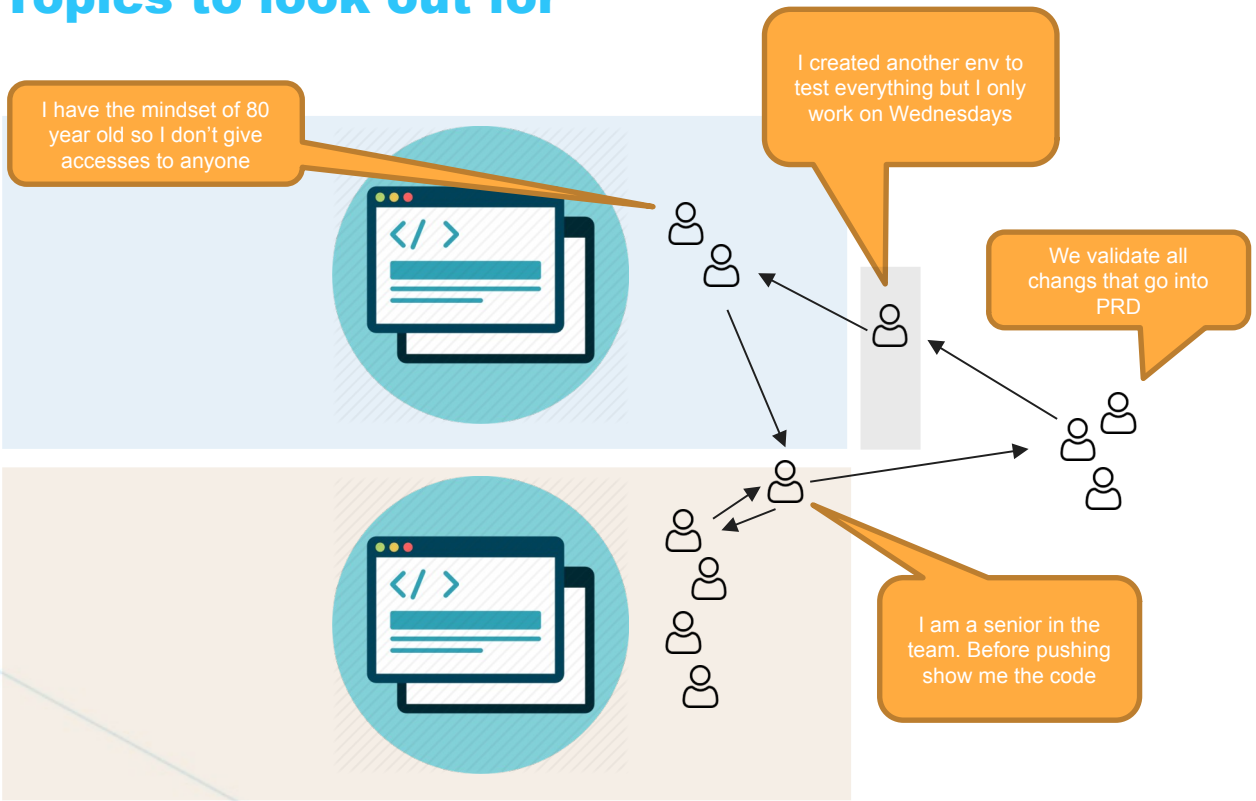
Topics to look out for





- Production Maintenance
- Complicated bureaucracy to deploy to production

Topics to look out for



- Production Maintenance
- Complicated bureaucracy to deploy to production

Topics to look out for



pandas v0.25.12
(no numpy)

PROD



pandas v2.04.15
numpy v1.0.4

DEV

- Production Maintenance
- Complicated bureaucracy to deploy to production
- Different infrastructure setup

IS THERE REALLY NO WAY TO SOLVE THIS LAST ONE?

Topics to look out for



pandas v0.25.12
(no numpy)

PROD



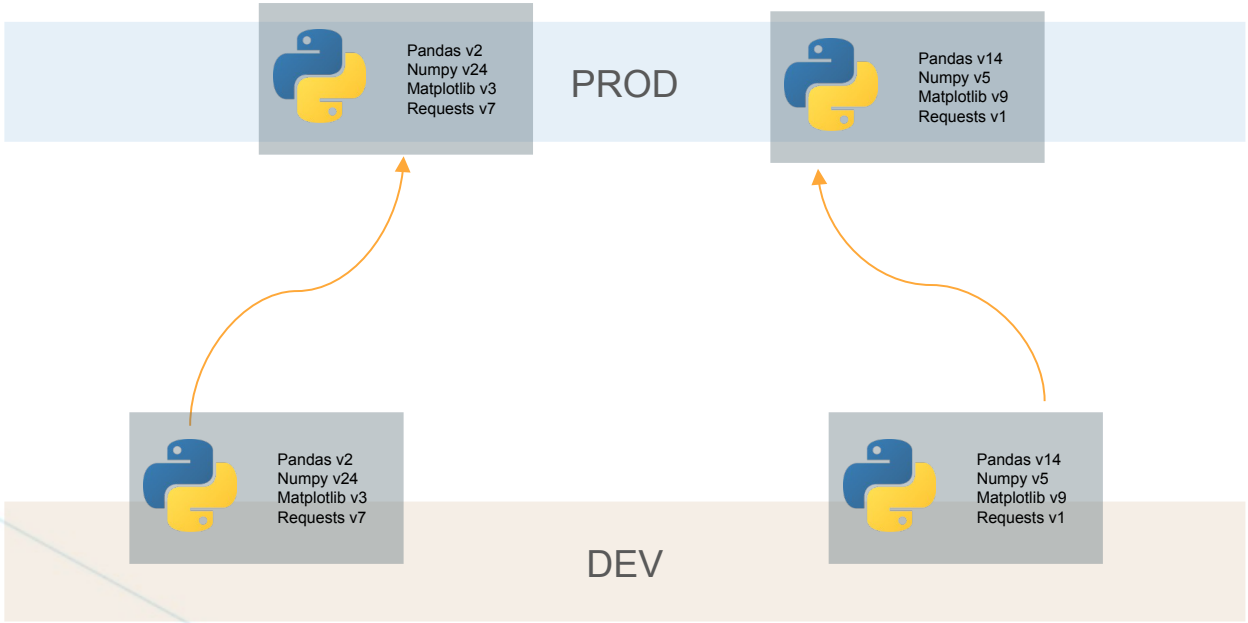
pandas v2.04.15
numpy v1.0.4

DEV

- Production Maintenance
- Complicated bureaucracy to deploy to production
- Different infrastructure setup

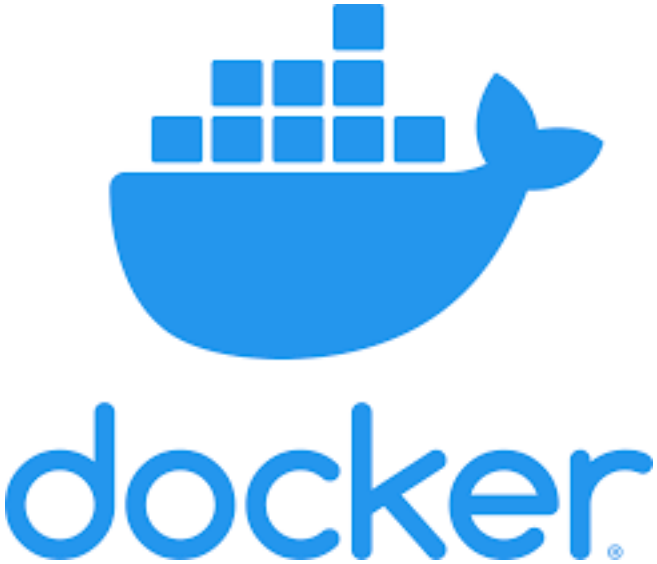
Virtual Environments

(venv)



”What about everything else besides Python packages:

SQL DB Setups, Drivers installed in Windows, etc?”


docker



Thank you